

Formalizing Data Structures Without Overdoing It: an Ontology for Dependency, Composition, and Value-Space Structure of Variables and Datasets

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Abstract

Datasets in scientific and data-intensive domains are rarely self-contained: they are derived from, aggregated over, and discretized from other data through processes whose semantics are typically left implicit in metadata catalogs. Existing vocabularies for statistical and observational data — RDF Data Cube, SSN/SOSA, QUDT, coverage standards — offer partial views of this problem, but none provides a general, domain-agnostic account of variables as abstractions over dimensional spaces, together with the dependency, specialization, and aggregation relationships that connect them, and the discretization processes (sampling, binning) that link continuous and discrete representations. This gap makes it difficult to reason automatically about dataset provenance, to relate datasets defined at different granularities, or to determine when two nominally different products represent the same underlying quantity. We present an ontology module that models variables and dimensional spaces in a way that is flexible enough to capture these relationships while keeping a pragmatic approach on the complexity of the representation. We describe the ontology’s classes and properties, discuss their relation to existing ontology design patterns and vocabularies, and illustrate the module through a worked example from the climate services domain, showing how a community-defined aggregating variable is progressively specialized into a concrete, finite, and serializable dataset.

Keywords

Ontology design pattern, Variable, Dimensional space, Dataset, Scientific data, Knowledge representation

1. Introduction

The increasing availability of environmental, scientific, and observational data has intensified the need for machine-actionable descriptions of datasets that go beyond simple metadata catalogs. In domains such as climate science, Earth observation, and numerical modelling, datasets are rarely monolithic objects: they are generated, transformed, aggregated, and re-derived from other datasets through complex processing pipelines, and the same underlying physical quantity (e.g., air temperature) may appear under many guises — as a raw model output or as a spatially or temporally aggregated product conforming to different community standards (e.g., a CMOR variable such as `tasmax`). Existing metadata standards and ontologies typically describe datasets as opaque, finite artifacts, characterized primarily by their access points, formats, and high-level thematic annotations. What they generally lack is a principled, semantically grounded account of the *variables* that a dataset represents, of the *spaces* in which those variables take values, and of the *derivation relationships* — dependency, aggregation, discretization — that connect a dataset to the other data objects from which it originates. This gap makes it difficult to automatically reason about dataset provenance, discover datasets that instantiate a more general variable at a different granularity, or determine whether nominally different datasets are compatible representations of the same quantity.

To address this need, we present an ontology module for the data-centric representation of variables and their dimensional spaces. The design is motivated by the observation, common to mathematics and to scientific data modelling more broadly, that a *variable* is fundamentally an abstraction whose

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meaning arises from its relations to other variables and from the association with a well-defined set of possible values — its *dimensional space*. Building on this intuition, the ontology aims to strike a balance between expressive flexibility, needed to capture the variety of ways variables can be defined, derived, and related, and well-defined formal semantics, needed to support automated reasoning and interoperability.

In the remainder of the paper, we detail the design of these classes and properties, illustrate their use through a worked example drawn from the climate services domain, and discuss how this representation supports reasoning tasks such as dataset discovery, provenance tracking, and the alignment of variables defined at different levels of aggregation or granularity.

1.1. Methodology and Notation

The ontology has been developed adopting the eXtreme Design methodology [1] in the context of the HACID (Hybrid Human Artificial Collective Intelligence in Open-Ended Domains) project ¹, for a use case involving the representation of climate-related datasets and variables. For the project, a common “upper level-like” ontology — hereafter the *top-level* ontology — was designed, of which the ontology presented here (the *data* ontology) is a specialization. The top-level ontology is an adaptation of the broadly adopted upper level ontology DOLCE Ultra Light (DUL) [2] and for present purposes can be considered equivalent to it. The mentioned ontologies are currently maintained in a public git repository² and are available online with persistent URIs, respectively at <https://w3id.org/hacid/onto/top-level> and <https://w3id.org/hacid/onto/data>. In this paper the namespaces of the corresponding terms will be abbreviated with the prefixes `top:` and `data:`. For the usage example other prefixes are used for individuals. Albeit most of them are precisely defined in the context of the HACID project, they are not specified here since they are not relevant to the purpose of documenting the ontology. The diagrams describing the ontology (figures 1 and 2) adopt the Graffoo visual notation [3]. The diagram illustrating the usage example (Figure 3) adopts the Chowlk notation [4], which allows for a more concise representation of instance data than Graffoo.

1.2. Competency Questions

Consistently with the eXtreme Design methodology, the design of the ontology has been driven by a set of competency questions (CQs). Since the module is intended to be domain-agnostic rather than tied to the climate services use case that originally motivated it (see Section 3), the CQs were elicited both from that use case and, deliberately, from scenarios in other data-intensive domains — official statistics, epidemiology, finance, remote sensing, energy, and social-science surveys — so as to probe whether the modelling patterns identified for climate data (dependency, specialization, rescaling, componenthood, aggregation, discretization, finiteness, provenance, equivalence) generalize beyond it. Table 1 reports a representative subset of the full CQ set, one per modelling pattern and spanning several domains; the complete list (37 CQs at the time of writing), together with the ontology terms each of them exercises, is published as supplementary material alongside the ontology sources³. As an illustration, the specialization pattern used in Section 3 to derive `mip:tasmax` from `cf:air_temperature` is not climate-specific: the same `data:isSpecializationOfVariable` property answers the analogous Epidemiology question in Table 1, obtained by constraining the geodetic-region independent variable instead.

2. Ontology Description

The module is organized around three interrelated groups of classes: (i) the **Variable** hierarchy, capturing dependency, derivation, and aggregation; (ii) the **DimensionalSpace** hierarchy, capturing the

¹<https://www.hacid-project.eu/>

²<https://github.com/hacid-project/knowledge-graph>

³<https://github.com/hacid-project/knowledge-graph/blob/main/ontologies/data/competency-questions.md>

A representative subset of the competency questions used to drive and validate the ontology design.

structure of value spaces and their discretization; and (iii) a lightweight **Dataset / DataSource / Process** module connecting the abstract variable model to concrete, serialized data and to the processes that produce or consume it. Each term of the ontology has an associated label (using `rdfs:label`) and extended description (using `rdfs:comment`) in English. All the properties have defined domain and ranges. Furthermore, all object properties are defined in direct-inverse pairs for symmetry (e.g., `data:dependsOnVariable` `data:hasDependentOnVariable`). For reasons of conciseness, generally only one of the two is mentioned in the following description.

All terms of the ontology presented here (the data ontology) are grounded in HACID top-level ontology classes and properties (top:Concept, top:Entity, top:InformationEntity, top:Region, top:Parameter, top:associatedWith, top:hasPart), which, being mostly equal to the DUL upper level ontology, favours direct alignment with DUL and derived ontologies and patterns.

```

graph TD
    data:FiniteVariable -- rdfs:subClassOf --> data:Variable
    data:DerivedVariable -- rdfs:subClassOf --> data:Variable
    data:DerivedVariable -- data:derivedFromVariable --> data:Variable
    data:Region -- data:hasSelectedRegion --> data:VariableSpecialization
    data:VariableSpecialization -- data:isSpecializedAccordingTo --> data:Variable
    data:VariableSpecialization -- data:aggregatesVariable --> data:Aggregation
    data:Aggregation -- data:definesAggregation --> data:Quantization
    data:Aggregation -- data:hasSuggestedQuantization --> data:Quantization
    data:Quantization -- data:hasDiscretization --> data:DiscreteDimensionalSpace
    data:IndependentVariable -- rdfs:subClassOf --> data:Variable
    data:DependentVariable -- rdfs:subClassOf --> data:Variable
    data:IndependentVariable -- data:dependsOnVariable --> data:Variable
    data:DependentVariable -- data:dependsOnVariable --> data:Variable
    data:IndependentVariable -- owl:disjointWith --> data:DependentVariable
    data:DimensionalSpace -- rdfs:subClassOf --> data:IndependentVariable
    data:DiscreteDimensionalSpace -- rdfs:subClassOf --> data:DimensionalSpace
    data:DiscreteDimensionalSpace -- data:isSpecializationOn --> data:IndependentVariable

```

Figure 1: Graffoo diagram depicting the part of ontology related to variables.

The class `data:Variable` (see Figure 1) is a top concept of the module, linked to its value space via `data:hasValuesOn` (domain `data:Variable`, range `data:DimensionalSpace`). Two disjoint subclasses, `data:IndependentVariable` and `data:DependentVariable`, capture the basic dichotomy in the role of a variable in a context. A `data:DependentVariable` is connected to the independent variables it depends on via `data:dependsOnVariable`.

Furthermore, the ontology models **derivation**, i.e. the fact that a variable (either dependent or independent) can be somehow *computed* from other variables (either dependent or independent), through the class `data:DerivedVariable` and the property `data:derivedFromVariable`. The notion of computation applies also to non finite (still undetermined) variables, in the same sense that one can say that circumference of a circle can be always computed from its radius, even if the values are unknown. It should be noted that, as described below, variable composition is treated as a specific case of this notion of computation we are adopting for the derivation relationship. Multiple specializations of derivation are distinguished as sub-properties of `derivedFromVariable`:

- `data:isSpecializationOfVariable`, a variable that specializes a more generic one by constraining one or more of the independent variables it depends on to a specific region of their dimensional space (e.g., “air temperature *near the surface*” as a specialization of “air temperature”)
- `data:isRescaledVersionOfVariable`, a variable obtained from another by a change of scale or unit, without changing its dependency structure;
- `data:isComponentVariableOf` (inverse `data:hasComponentVariable`), a variable that is a structural part of a composite one (e.g., latitude and longitude as components of a geodetic-coordinate variable).

The shortcut property `data:holdsSpecializationOfVariable`, which is a common super property of `data:isSpecializationOfVariable` and `data:hasComponentVariable`, is used to represent (optional) specialization followed by (optional) composition, when it is inconvenient to define an intermediate node for the purpose. A symmetric property `data:isEquivalentTo` allows asserting that two variables — however differently derived — always coincide in value, supporting alignment between independently modelled variable graphs (e.g., across different data providers).

Aggregation is modelled without committing to a specific granularity: a `data:DependentVariable` is *aggregating* if it is linked, via `data:definesAggregation`, to one or more instances of `data:Aggregation`. Each `data:Aggregation` is associated with exactly the variable being grouped over (`data:aggregatesVariable`, which must be one of the variables the dependent variable `dependsOnVariable`) and, optionally, with one or more suggested granularities via `data:hasSuggestedQuantization`, pointing to candidate `data:Quantization` instances. This indirection — aggregation *function* versus aggregation *grid* — is what allows a single aggregating variable (e.g., a CMOR `tasmax`-like construct) to remain agnostic about whether it will ultimately be realized at daily, monthly, or seasonal granularity.

Finally, `data:FiniteVariable` (aliased to `data:Dataset` via `owl:equivalentClass`) singles out variables whose value set is finite. The definition of a dataset as a finite variable is consistent with the data oriented perspective adopted for this ontology. For comparison, it is similar to the concept of `qb:Dataset` in the RDF Data Cube Vocabulary (QB) [5], while the `dcat:Dataset` of the Data Catalog Vocabulary (DCAT) [6], conversely, is focused on publication/organisational aspects. From a modelling point of view, this choice enables directly reusing for datasets the properties that have been defined for variables — rescaling, component-hood, specialization — which connect datasets to other variables (that may or may not be datasets) with the same meaning they have on general (so not necessarily finite) variables.

2.3. Dimensional spaces and quantization

`data:DimensionalSpace` is a subclass of `data:IndependentVariable`, carrying a self-restriction on `hasValuesOn` (`owl:hasSelf true`): every dimensional space is, by construction, the default independent variable of itself. This design choice is what allows a dimensional space to be used directly wherever a variable is expected, rather than requiring an explicit auxiliary variable.

Structurally, a dimensional space is composed of `data:Region` instances, exposed via `data:hasRegion`, and may expose specific `data:Points` via `data:hasPoint`, optionally anchored by a distinguished `data:hasReferencePoint` (e.g., the North Pole for geodetic coordinates). Bounding is modelled at two levels of precision: `data:hasExactBoundingRegion` for regions that match the space's extension exactly, and `data:hasApproximateBoundingRegion` for regions that approximate it within an unspecified error, both sub-properties of the generic `data:hasBoundingRegion` and both aligned to corresponding exact/approximate distinctions inherited from the top-level ontology (`top:hasExactRegion/top:hasApproximateRegion`).

The primary split is between `data:Continuum` and `data:DiscreteDimensionalSpace` (asserted disjoint), orthogonal to the split between `data:BoundedDimensionalSpace` (finite extent, not necessarily finite or even countable cardinality) and unbounded spaces. The intersection between classes `data:DiscreteDimensionalSpace` and `data:BoundedDimensionalSpace` is the `data:FiniteDimensionalSpace` class, which corresponds to spaces that can be fully enumerated.

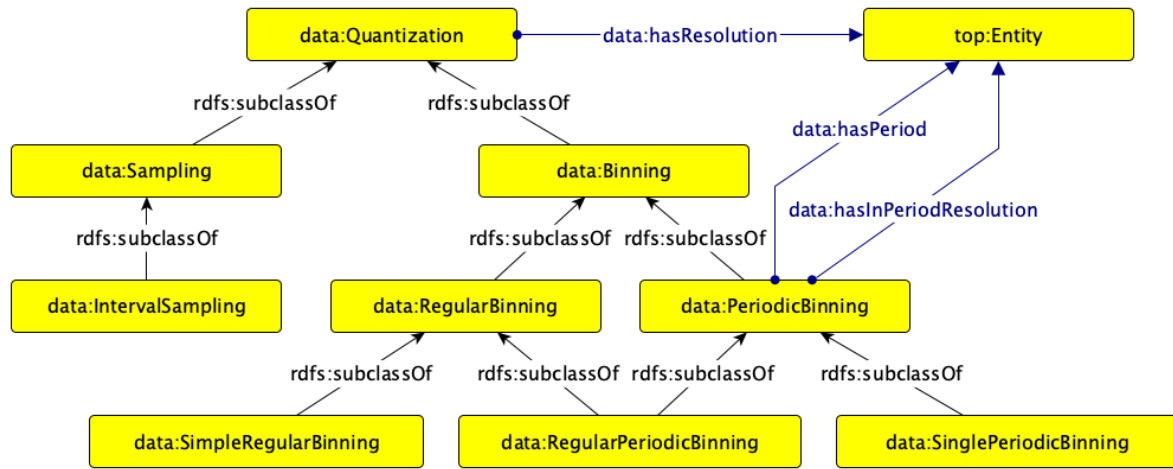


Figure 2: Graffoo diagram depicting the part of ontology related to quantizations.

Discretization of a continuum is captured by `data:Quantization`, itself a `DiscreteDimensionalSpace`, connected to the continuum it discretizes via `data:hasDiscretization`, a sub-property of the more general `data:basedOnDimensionalSpace` (which in turn specializes `isSpecializationOfVariable`, treating “being a discretization of” as one particular way a dimensional space can specialize another). Every Quantization (see Figure 2) carries a `data:hasResolution` (with exact/approximate variants `hasExactResolution/hasApproximateResolution`) and an optional `data:hasOffset` relative to the reference frame. Two quantization strategies are distinguished:

- `data:Sampling`, where discretized points correspond to individual sampled points of the continuum, specialized as `data:IntervalSampling` for regularly spaced samples (e.g., a time instant every 3 hours);
- `data:Binning`, where discretized points correspond to sets/regions of the continuum (e.g., $10^\circ \times 10^\circ$ grid cells on the geodetic surface).

`data:Binning` is further specialized in two (partially overlapping) sub-cases:

- `data:RegularBinning` models bins of uniform size, further specialized as `data:SimpleRegularBinning` (e.g., simple monthly bins over time);
- `data:PeriodicBinning` models bins that repeat over a cycle, exposing `data:hasPeriod` and `data:hasInPeriodResolution` (the resolution adopted *within* each period), further specialized as `data:SinglePeriodicBinning` and `data:RegularPeriodicBinning` (e.g., 12 monthly bins aggregated over the whole time extent, or over successive 20-year intervals, respectively).

Domain-specific extensions of `Region` and `Point` are provided for the two dimensions most relevant to the climate-services use case: `data:GeodeticRegion/data:GeodeticPoint`, aligned with `geosparql:Geometry` for interoperability with GeoSPARQL-based spatial tooling, and `data:TemporalRegion`, equipped with datatype properties `data:hasStartDateTime/data:hasEndDateTime`. Compositional structure across dimensions is expressed via `data:hasSubDimensionalSpace` — itself a sub-property of `hasComponentVariable` — allowing, e.g., a purely temporal or purely spatial dimensional space to be modelled as a projection of a composite spatio-temporal one.

2.4. Datasets, data sources, and provenance

Because `data:Dataset` is equivalent to `data:FiniteVariable`, the dataset-level vocabulary is deliberately minimal and reuses the variable machinery already described rather than duplicating it. A `data:Dataset` is connected to one or more concrete serializations through `data:hasDataSerialization`, whose range is `data:DataSource`, an `top:InformationEntity` representing a location (typically a URL, exposed via the datatype property `data:hasURL`) together with the technical means to access the data. Each `data:DataSource` is in turn associated with one or more supported `data:DataFormat` instances (e.g., NetCDF, CSV) via `data:hasAvailableDataFormat`, with the ontology explicitly recommending the reuse of external controlled vocabularies for concrete format identifiers rather than minting new ones.

Provenance is handled by a minimal process model: `data:DataGeneratingProcess` produces one or more `Datasets` (`data:hasOutput`), `data:DataConsumingProcess` consumes one or more `data:Datasets` as input (`data:hasInput`), and `data:DataTransformation` is defined as the intersection of the two, modelling processes — such as an aggregation or reformatting pipeline — that both consume and generate datasets. This ontology facet is intentionally lightweight and is designed to be aligned with, rather than substitute for, general-purpose provenance vocabularies such as PROV-O, with `data:DataGeneratingProcess/data:DataConsumingProcess` playing a role analogous to `prov:Activity` restricted to the data-generation/consumption facet.

3. Usage Example

To ground the abstractions introduced above, we describe a small but representative example, taken from the climate services domain that originally motivated this work: the publication of a dataset of *average daily maximum near-surface air temperature over the whole of Italy for the year 2024*, derived from some process (possibly including observations, simulations, aggregations, etc.) — i.e., a dataset with a single value per day for the whole region (rather than, e.g., a spatial grid), having one independent variable (time) and one dependent variable. Despite its apparent simplicity, this example touches on every main modelling mechanism offered by the ontology — dependency, specialization, aggregation, discretization, finiteness, and provenance — and can be represented compactly as the instance graph shown in Figure 3.

Generic and specialized variables. At the top of the graph, instances of `data:Continuum` are used to represent physical dimensions relevant to the case: `dim:time`, `dim:space`, `dim:temperature`. The standard variable `cf:air_temperature` is represented as a `data:DependentVariable` with values on `dim:temperature`, depending (`data:dependsOnVariable`) on two variables (`dim:space` and `dim:time`). The variable `mip:tasmax`, representing *near-surface* air temperature, is linked to it via `data:isSpecializationOfVariable`: conceptually, `tasmax` is the same physical quantity as `air_temperature`, restricted to a specific vertical region (near the surface), without further committing to any particular time range, spatial extent (on the surface), or granularity. It still depends on `dim:time` but it does not depend on `dim:space` anymore. As the vertical (earth-bound) spatial component has been fixed, it now depends on `dim:geodetic`, the sub-dimensional space of `dim:space` that represents coordinates on the geodetic surface (e.g., standard latitude and longitude).

Aggregating variable. Furthermore, `mip:tasmax` is *aggregating* in the geodetic surface and time.

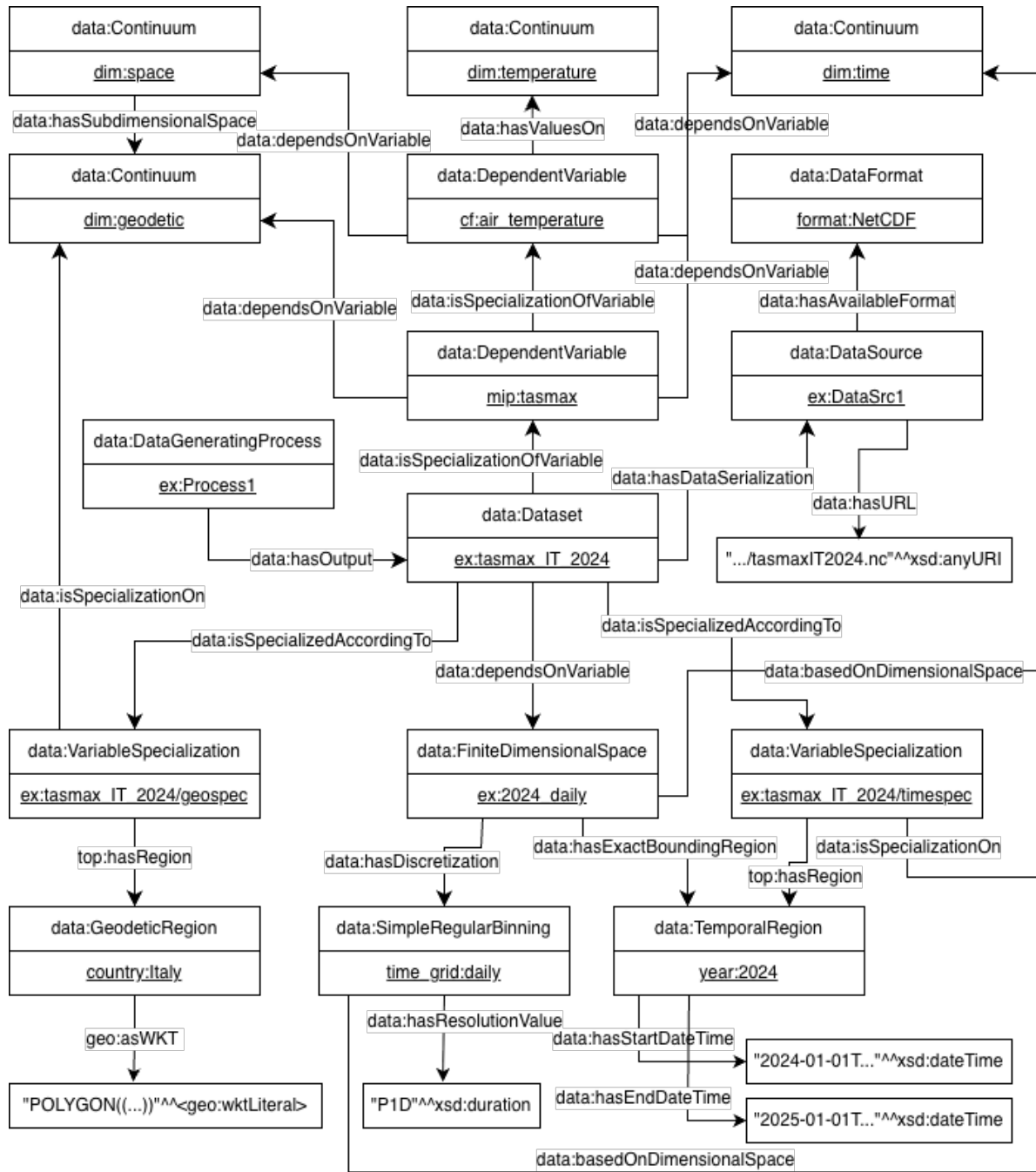


Figure 3: Chowlk diagram depicting the usage example.

This means that is associated with specific aggregation functions along the two axes⁴ This can be explicitly represented by linking the variable via `data:definesAggregation` to two `data:Aggregation` instance, one for each dimension. This last bit is not shown in the diagram for conciseness. At this stage, `tasmax` still denotes an *unbounded, ungrounded* aggregating variable: it specifies *how* values are to be combined, but not *over which* concrete spatiotemporal span or granularity.

Concrete grid. The instance `ex:2024_daily` is a `data:FiniteDimensionalSpace` obtained by applying the discretization `time_grid:daily` (via `data:hasDiscretization`) to `dim:time` (via `data:basedOnDimensionalSpace`), bounded to the year 2024 through `data:hasExactBoundingRegion`, point-

⁴Along time it is computed using the maximum up to the day granularity and then (e.g., month, year) using the average; along space it is computed using average.

ing to a `data:TemporalRegion` with the corresponding start and end date-times. This method can be adopted also to represent spatial grids (which in this simple case are not needed) as well as quantizations over arbitrary dimensional spaces.

Coverage. The coverage of some independent variable y (in respect to dependent variables, typically space and time) can be represented in two ways: as bounding region of the corresponding (finite) dependent variable, as shown above; as specialization over a dependent variable of a generalization of y . The latter method is showcased in the example by the two instances of `data:VariableSpecialization`, `ex:tasmax_IT_2024/timespec` and `ex:tasmax_IT_2024/geospec`, that represent the coverage respectively over time and geodetic surface (year 2024 and Italy). It should be noted that the “specialization” method is the only one available to specify the coverage with respect to some variable that has collapsed and is not available at that level (like geodetic surface in this case).

The dataset. The variable `ex:tasmax_IT_2024` represents the concrete, publishable product: it is linked via `data:isSpecializationOfVariable` to `tasmax`, now dependsOnVariable the finite grid `ex:2024_daily` instead of the unbounded continua, and has coverage defined by the specializations `ex:tasmax_IT_2024/timespec` and `ex:tasmax_IT_2024/geospec`. Because the only variable it depends on is finite, `ex:tasmax_IT_2024` satisfies the definition of `data:FiniteVariable` and is therefore, by the owl:equivalentClass axiom, a `data:Dataset` — formally justifying its treatment as a dataset rather than as an abstract variable.

Serialization and provenance. The dataset is linked, via `data:hasDataSerialization`, to a `data:DataSource` instance exposing its URL and, via `data:hasAvailableDataFormat`, the NetCDF `data:DataFormat` in which it is available. Finally, `ex:Process1`, an instance of `data:DataGeneratingProcess`, is connected to the dataset via `data:hasOutput`, recording that the dataset was produced by a specific (here unspecified) process; were the same process also to consume an input dataset, it would be typed as a `data:DataTransformation`, using `data:hasInput` in addition to `data:hasOutput`.

Read end-to-end, the example illustrates the intended usage pattern of the ontology: a chain of increasingly specific variables — general quantity, named community variable, aggregating variable, concretely gridded dataset — connected by explicit, differently-typed derivation links (`data:isSpecializationOfVariable`, dependency, aggregation).

4. Related Work

The representation of statistical and scientific data has been the subject of substantial work in the Semantic Web community, spanning general-purpose vocabularies for multidimensional data, ontologies for observations and measurements, and reusable ontology design patterns; we organize the discussion along these lines.

4.1. Statistical and multidimensional data vocabularies

The RDF Data Cube Vocabulary (QB), based on the multidimensional model used in the Statistical Data and Metadata eXchange (SDMX) standard for exchanging statistical data, represents observations as points in a space defined by dimensions, measures, and attributes [5]. While QB is widely adopted for publishing statistical datasets as Linked Data, its dimensional model is comparatively coarse: dimensions are treated as code lists or simple value spaces, with no explicit account of continuous versus discrete spaces, discretization processes, or the derivation relationships between variables that are central to our module. Extensions such as QB4ST (RDF Data Cube extensions for spatio-temporal components) [7] and QB4OLAP [8] add spatial, temporal, and OLAP-style hierarchical structure to data cubes, but remain focused on the observation-as-point-in-a-cube paradigm rather than on a general theory of variables and their dimensional spaces. The DDI-RDF Discovery Vocabulary [9], derived from the Data Documentation Initiative, similarly targets the discovery and description of statistical and survey datasets, emphasizing variable-level metadata (e.g., concept, unit, representation) but without a formal treatment of dependency, aggregation, or discretization.

4.2. Observations, measurements, and sensor data

The SSN/SOSA family of ontologies (Semantic Sensor Network Ontology and its lightweight core, Sensor, Observation, Sample, and Actuator) [10] and the related ISO 19156 Observations and Measurements (O&M) standard [11] model an observation as an act that estimates the value of a property of a feature of interest using some procedure. These ontologies provide a rich vocabulary for provenance of individual observations but treat the value space of a property largely as an opaque range, delegating the description of units and quantity kinds to companion vocabularies such as QUDT or the OM ontology of units of measure (see below). They do not address the compositional structure of dimensional spaces (e.g., discretizations of a continuum via sampling or binning) nor the notion of a variable that is itself derived, through aggregation, from another variable, both central concerns of our module. The Extensible Observation Ontology (OBOE) [12], developed in the ecological sciences, is closer in spirit in that it explicitly models measurements as characterizing entities via a “characteristic” and associated value, and supports derived (computed) observations; however, it does not formalize the underlying value space as a first-class, potentially continuous or discretized structure.

4.3. Units, quantities, and coverage data

QUDT (Quantities, Units, Dimensions and Types) and the closely related OM ontology [13] provide detailed vocabularies for quantity kinds, units, and dimensions, and are commonly used in conjunction with observation ontologies to type the values a variable can take. These vocabularies are complementary to, rather than overlapping with, our module: they describe *what kind* of quantity a value represents, while our `data:DimensionalSpace` hierarchy describes the *structure* of the space of values (continuous, discrete, sampled, binned) independently of its physical quantity kind. In the geospatial and coverage-data community, the OGC CoverageJSON [14] and GML/Coverage Implementation Schema [15], together with GeoSPARQL [16, 17] for spatial data, address the discretization of continuous fields into grids, but tie discretization tightly to spatial coordinate reference systems rather than treating it as a general property applicable to any dimensional space (temporal, spatial, or otherwise), as our `data:Quantization`, `data:Sampling`, and `data:Binning` classes do.

4.4. Provenance and data-generating processes

The PROV-O ontology [18], the W3C standard for representing provenance, offers general-purpose classes for activities, entities, and agents, and has been widely reused to describe how datasets are generated and transformed. Our `data:DataGeneratingProcess` and `data:DataTransformation` classes are designed to be compatible with, and can be aligned to, PROV-O’s `Activity` and `Generation/Usage` relations; however, PROV-O itself is agnostic to the internal structure of the data being generated or consumed, which is precisely the gap our module fills by connecting provenance-level processes to the variable- and dimensional-space-level description of their inputs and outputs.

4.5. Ontology design patterns

Our design draws on established content ontology design patterns (ODPs), notably the *Region* pattern, used to model qualitative and quantitative regions as first-class entities rather than as bare literal values, and patterns for *parameter* and *observable property* reuse from the ODP portal. The choice to model a `data:DimensionalSpace` as composed of `data:Region` instances, and to allow a dimensional space to double as an `data:IndependentVariable` when no ambiguity arises, follows the general ODP principle of minimizing modelling overhead in the common case while preserving the possibility of more explicit representation when needed. This is in line with recommendations from the broader ontology engineering literature on balancing expressiveness and usability in modular, reusable ontology modules. Taken together, the vocabularies surveyed above each address one facet of the problem, but none provides a unified, domain-agnostic account of variables as abstractions over dimensional spaces together with their dependency, aggregation, and discretization relationships; the module presented in

this paper is intended to fill this gap with a lightweight core that can be aligned with and extended by these vocabularies.

5. Conclusions

We presented an ontology module for the representation of variables and their dimensional spaces, filling a gap left by existing statistical, observational, and coverage-data vocabularies: a domain-agnostic account of how variables relate to one another — through dependency, specialization, rescaling, componenthood, and aggregation — and of how their value spaces are structured, bounded, and discretized.

Three design choices, in particular, characterize the module and are offered as patterns that may be of independent interest to the ontology design and patterns community: treating a dimensional space as its own default independent variable, so that the space/variable distinction can be collapsed exactly where it adds no information; decoupling an aggregation’s function from the grid over which it is applied, so that community-defined aggregating variables can remain agnostic to granularity until a concrete quantization is chosen; and factoring the generic notion of variable derivation into distinguishable, independently query-able sub-relations — specialization, rescaling, componenthood, discretization — rather than a single undifferentiated “derived from” link. The worked example demonstrated how these patterns combine in practice, tracing a single physical quantity from a generic community-defined variable to a concrete, finite, publishable dataset, complete with its serialization and generating process. The module has been applied to climate and geospatial data, on which the example is based; we have started evaluating its coverage and reusability against a broader range of scientific domains through the competency questions discussed in Section 1.2, spanning statistics, epidemiology, finance, remote sensing, and energy data among others.

The module is intentionally narrow in scope: it does not attempt to duplicate the roles played by established vocabularies for units and quantity kinds, sensor and observation provenance, or general-purpose activity tracking, relying instead on alignment with resources such as QUDT/OM, SSN/SOSA, GeoSPARQL, and PROV-O for these concerns. This narrowness is deliberate, and we regard it as a strength for reuse, but it does leave open several directions for future work. As an example, the current treatment of aggregation is qualitative: a `data:Aggregation` records that a function is applied along a variable, and optionally suggests a granularity, but does not formalize the aggregation function itself (e.g., distinguishing mean from maximum in a machine-actionable way); integrating a small taxonomy of aggregation functions, or alignment with existing function ontologies, would strengthen this part of the model. Furthermore, while we have started evaluating in-house its coverage and reusability beyond the climate data domain, this aspect needs to be improved by evaluating the module against existing datasets from these domains and assessing how well it supports discovery and provenance-tracking tasks in practice.

We believe that by isolating variable and dimensional-space-level structure as a distinct, reusable concern — separate from statistical cube structure, sensor provenance, or unit typing — this module can serve as a connective layer between the several, otherwise disconnected, vocabularies that currently address adjacent facets of scientific data description; and we hope its patterns prove useful beyond the climate services use case that originally motivated them.

Declaration on Generative AI

During the preparation of this work, the author(s) used Claude Sonnet 5 for the manuscript production (not for the actual ontology design) in order to: Drafting content. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication’s content.

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